

The Fibonacci–Mellin Transform: Vandermonde Structure, Non-Isometry, and Finite-Field Orthogonality

Pasquale Camelia*

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Abstract

The Discrete Fourier Transform resolves the additive, periodic structure of a signal. For signals whose structure is instead multiplicative, log-periodic, or discretely scale-invariant, the natural tool is a Mellin transform on a multiplicative lattice. This paper introduces the Fibonacci–Mellin Transform (FMT), a finite Vandermonde frame sampled on a golden-ratio scale lattice, and studies its algebraic and representational structure. The transform output splits naturally into a two-sheet LOW/HIGH decomposition whose ratio probes scale structure and is invariant to global gain. We establish three structural facts: the frame is non-unitary, with a large isometry deficit that is the signature of its projective character; no real diagonal reweighting can unitarize it at the size studied; and its polar factor provides an associated unitary readout. From the two sheets we define a set of scale-coherence diagnostics and show numerically that they separate golden-structured signals (log-periodic, scale-drift, Fibonacci-driven) from harmonic controls. The FMT is not proposed as a replacement for Fourier analysis, as a faster transform, or as a compression tool; it is a fixed, parameter-free scale-transfer frame whose value is gain-invariant scale readout. An applied companion study uses this readout as a validation gate for acoustic-emission monitoring.

Keywords. Fibonacci–Mellin transform; scale-memory analysis; two-sheet decomposition; non-unitary frame; polar unitarization; scale-coherence diagnostic; gain-invariant readout.

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*pasquale.camelia@gmail.com. ORCID: 0009-0006-4779-4647. © 2026 Pasquale Camelia. Licensed under CC BY-NC 4.0 — free to share and cite with attribution; commercial use prohibited.

1 Introduction

1.1 Scale memory vs frequency memory

The DFT on $\mathbb{Z}_N$ answers the question: *which harmonic frequency is present?* It operates on a closed additive cycle $k \mapsto k + N \equiv k$ and resolves scalar $U(1)$ characters $e^{-2\pi i j k / N}$. Each bin $X_j \in \mathbb{C}$ carries a single phase and a single energy value.

A different question arises when a signal's structure is organised multiplicatively rather than additively: *where is the signal inside a φ -autosimilar scale hierarchy, and how did it advance from one scale level to the next?* The golden fine lattice $x_k = \varphi^{k/5}$ closes quasi-periodically:

$$k \mapsto k + 10 \implies x_{k+10} = \varphi^2 x_k \neq x_k. \quad (1)$$

After one full traversal of the 10-node lattice, the scale advances by $\varphi^2 \approx 2.618$ rather than returning to the start. This advance is the *scale memory* that the FMT preserves and converts into measurable phase-coherence features.

1.2 Signal classes where the FMT is relevant

Table 1 classifies signal classes by their suitability for FMT analysis. The FMT is useful when the signal's structure is multiplicative, log-periodic, self-affine, or discretely scale-invariant; it offers no advantage over the DFT for purely harmonic signals.

1.3 Motivating experiments

Dumitrescu et al. [2] realized a dynamical symmetry-protected topological phase (ED-SPT) by driving 10 ytterbium qubits with a Fibonacci pulse sequence. Xu et al. [3] demonstrated non-abelian braiding of Fibonacci anyons on a superconducting processor. These experiments motivate two distinct candidate domains: Fibonacci-timed drives in EDSPT systems, and Fibonacci-category braid structures. The present paper does not claim a direct validation on either dataset; this is left as open problem P3.

1.4 Main results

- (i) *LOW/HIGH two-sheet decomposition* (Theorem 3.1, §3): the raw FMT has $y_j = \text{LOW}_j$ and $y_{j+5} = \text{HIGH}_j$, where the HIGH row is the LOW row pointwise Artin-twisted before projection: $(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{j,k}$. GP(5,2) models the two-sheet indexing. CT(2,2) is optional post-processing, not part of $\mathcal{F}_\varphi$.
- (ii) *$U(2)/SU(2)$ doublet structure* (§4): the doublet $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$ carries a $U(1) \times SU(2)$ -type representation; Stokes-like observables (S_x, S_y, S_z) provide scale-internal feature vectors for each frequency index j .

Table 1: Domain applicability of FMT scale-memory analysis. *High*: FMT directly resolves the scale structure. *Medium*: FMT provides complementary scale-coherence features. *Low*: DFT is sufficient and optimal.

Signal class	Priority	Structural reason
Log-periodic signals	High	Oscillations at scales φ^m ; Mellin-native
Discrete scale invariance	High	$f(\lambda x) = \lambda^h f(x)$ for discrete λ near φ
Quasi-periodic drives	High	Fibonacci pulse sequences; φ -rational frequency support
OMA/SHM modal families	Candidate	Useful if diagnostic target is log-scale coherence loss; validation against MAC and SSI required
SAR/DEM fractal texture	Candidate	Plausible for self-affine textures; no benchmark in this paper
$1/f^\alpha$ noise	Medium	Scale coherence complements spectral slope estimate
Multifractal turbulence	Medium	FMT as compact scale-ordering feature
NDVI / vegetation texture	Medium	Hierarchical canopy structure; log-scale patterns
Pure harmonic signals	Low	DFT is optimal; FMT offers no sparsity advantage
Standard OFDM / QAM	Low	Additive periodic closure; DFT is the native transform

- (iii) *φ -memory cell* (Proposition 6.1, §6): $G_j(k) \in U(2)$ is index-dependent. No rational commensurability of $\delta_{j,k}$ with π is detected numerically on the tested lattice phases; the observed phase orbits are consistent with quasi-periodicity ($G_j^n \neq I$ for all tested n).
- (iv) *Diagonal unitarization obstruction* (Proposition 8.1, §8): no real diagonal weights unitarize $\mathcal{F}_\varphi$, numerically certified for $N = 10$.
- (v) *Polar unitary readout* (Theorem 9.2, §9): $U_\varphi = WV^* \in U(10)$, embeds into $U(16)$, admits canonical POVM and Halmos block-encoding.
- (vi) *Scale-coherence algorithm* (Algorithm 1, §7): a complete seven-step pipeline from signal to feature vector.

What is not claimed. The FMT does not outperform the DFT in spectral sparsity: $G_{\text{FMT}} < G_{\text{DFT}}$ (Gini) in all benchmarks, because the 4.70° Vandermonde arc makes FMT basis functions nearly parallel. $R_{\text{FMT}}^2 \approx 1$ for harmonic signals too; the operative discriminant is the contrast ΔTV and ΔR^2 , not R_{FMT}^2 alone.

2 The FMT and the Golden Lattice

Definition 2.1 (Golden fine lattice). $x_k = \varphi^{k/5}$, $k = 0, \dots, 9$; log-space steps $u_k = (k/5) \log \varphi$; quasi-periodic closure $x_{10} = \varphi^2 \neq 1$.

Definition 2.2 (Fibonacci–Mellin Transform).

$$(\mathcal{F}_\varphi)_{jk} = \frac{1}{\sqrt{10}} e^{i\omega_j u_k}, \quad \omega_j = \varphi^{(j-5)/5}, \quad j, k = 0, \dots, 9. \quad (2)$$

The 10 Mellin frequencies ω_j form a geometric sequence with ratio $\varphi^{1/5}$, centred at $\omega_5 = 1$. Writing $z_j = e^{i\theta_j}$ with $\theta_j = \omega_j \log \varphi / 5$, $\mathcal{F}_\varphi$ is a Vandermonde matrix $(\mathcal{F}_\varphi)_{jk} = z_j^k / \sqrt{10}$; the 10 nodes z_j lie in an arc of 4.70° on S^1 .

Table 2: Structural comparison: FMT vs DFT.

Property	DFT on $\mathbb{Z}_{10}$	FMT on $\{x_k = \varphi^{k/5}\}$
Lattice	additive $\mathbb{Z}_{10}$	multiplicative $\{\varphi^{k/5}\}$
Closure	$\omega^{10} = 1$	$x_{10} = \varphi^2 \neq 1$
Kernel	$e^{-2\pi i j k / N}$ (Fourier)	$e^{i\omega_j u_k}$ (Mellin)
Indexing graph	$\mathbb{Z}_{10}$ (cycle)	$\text{GP}(5,2) \simeq \text{Cayley graph of } D_5$
Output structure	N scalar bins $X_j \in \mathbb{C}$	5 doublets $\Psi_j \in \mathbb{C}^2$
Representation	scalar $U(1)$	$U(1) \times SU(2)$ -type
Twiddle phase	$2\pi k / N \in \mathbb{Q}\pi$	$\delta_{j,k} / \pi$: no rational commensurability detected
Commutativity	$rs = sr$	$rs \neq sr$: non-comm. graph models sheet separation
Role	frequency memory	scale memory

Proposition 2.3 (Isometry deficit). $\sigma_{\max}(\mathcal{F}_\varphi) \approx 3.1534$, $\sigma_{\min} \approx 6 \times 10^{-18}$, $\kappa(\mathcal{F}_\varphi) \approx 5 \times 10^{17}$, $\|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F = 9.43$.

The 4.70° phase concentration and the isometry deficit are the same geometric fact: the Vandermonde nodes are nearly collinear on S^1 , so the FMT frame cannot distribute energy uniformly in the Euclidean metric.

3 LOW/HIGH Two-Sheet Decomposition

3.1 Mellin strand and Artin twist

The Mellin frequencies satisfy the scale-doubling relation $\omega_{j+5} = \varphi \omega_j$ for $j = 0, \dots, 4$. This separates the 10 frequencies into two sheets of 5. Define, for $(j, k) \in \{0, \dots, 4\} \times \{0, \dots, 9\}$:

$$R_j(k) = e^{i\omega_j u_k} \quad (\text{Mellin strand}), \quad A_j(k) = e^{i\omega_j \varphi^{-1} u_k} \quad (\text{Artin twist}). \quad (3)$$

The Fibonacci identity $1 + \varphi^{-1} = \varphi$ (from $\varphi^2 = \varphi + 1$) gives the fundamental relation:

$$\boxed{A_j(k) \cdot R_j(k) = e^{i\omega_j(1+\varphi^{-1})u_k} = e^{i\omega_j \varphi u_k} = R_{j+5}(k)}. \quad (4)$$

The HIGH strand R_{j+5} is the LOW strand R_j twisted by A_j . This is the algebraic core of the entire paper.

Theorem 3.1 (LOW/HIGH two-sheet decomposition). *For $f \in \mathbb{C}^{10}$ and $y = \mathcal{F}_\varphi f$, define*

$$\text{LOW}_j = \frac{1}{\sqrt{10}} \sum_{k=0}^9 f[k] R_j(k) = y_j, \quad (5)$$

$$\text{HIGH}_j = \frac{1}{\sqrt{10}} \sum_{k=0}^9 f[k] A_j(k) R_j(k) = y_{j+5}. \quad (6)$$

The raw FMT decomposes into a LOW sheet $(y_0, \dots, y_4)$ and a HIGH sheet $(y_5, \dots, y_9)$, related entry-wise by

$$(\mathcal{F}_\varphi)_{j+5,k} = A_j(k) (\mathcal{F}_\varphi)_{j,k}, \quad (7)$$

so that HIGH_j is the φ -scaled replica of LOW_j via the Artin twist.

Proof. $(\mathcal{F}_\varphi)_{jk} = R_j(k)/\sqrt{10}$ by definition. By (4), $R_{j+5}(k) = A_j(k)R_j(k)$, hence $(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{jk}$ entry-wise. Summing over k : $y_{j+5} = \frac{1}{\sqrt{10}} \sum_k f[k] A_j(k) R_j(k) = \text{HIGH}_j$. The Artin twist acts *inside* the sum, not on the already-projected coefficient LOW_j ; the two sheets are related at the matrix-entry level, not as a scalar multiplication of the output coefficients. $\square$

Remark 3.2 (CT(2, 2) is optional post-processing). Applying the Hadamard to the pair (y_j, y_{j+5}) produces *new combination channels*:

$$c_j^+ = y_j + y_{j+5}, \quad c_j^- = y_j - y_{j+5}. \quad (8)$$

These symmetric/antisymmetric combinations are not the raw FMT coefficients; they are an optional two-channel readout. Whether to apply CT(2, 2) is a design choice: the combination channels enforce the Hilbert Re/Im split of the rational filter $H(z) = P_4(z)/(1 - \alpha z)$ and additive periodicity $\omega^2 = 1$ on the boundary.

3.2 GP(5,2) as graph model of the two-sheet indexing

Remark 3.3 (GP(5,2) encodes the sheet structure). The generalized Petersen graph GP(5,2) is the Cayley graph of D_5 with generators r (rotation) and s (reflection), satisfying $srs = r^{-1}$ ($|D_5| = 10$). In the FMT:

- *Outer pentagon* $\{r^k, k = 0, \dots, 4\}$: the 5 LOW-sheet Mellin frequencies $\omega_0, \dots, \omega_4$; directed clockwise edges $r^k \rightarrow r^{k+1}$ represent the monotone frequency progression $\omega_j \rightarrow \omega_{j+1} = \varphi^{1/5}\omega_j$.
- *Inner pentagon* $\{sr^k, k = 0, \dots, 4\}$: the 5 HIGH-sheet frequencies $\omega_5, \dots, \omega_9 = \varphi\omega_0, \dots, \varphi\omega_4$; directed *anti-clockwise* edges because $srs = r^{-1}$ reverses the cycle orientation on the inner pentagon.
- *Radial edges* $r^k \leftrightarrow sr^{\sigma(k)}$ (where $\sigma = (0, 2, 4, 1, 3)$ is the Petersen skip): the reflection s realises the sheet coupling; each radial edge corresponds to an Artin twist $A_j(k)$.

The *abelian limit* $rs = sr$ collapses inner and outer pentagons to coincide: $A_j \equiv 1$, HIGH = LOW, two-sheet separation vanishes. The non-commutative graph structure *models* the sheet separation: when $rs \neq sr$, the radial edges carry a non-trivial Artin twist, and the two sheets become distinct.

Figure 1 shows the 5×2 butterfly; Figure 2 shows the GP(5,2) graph.

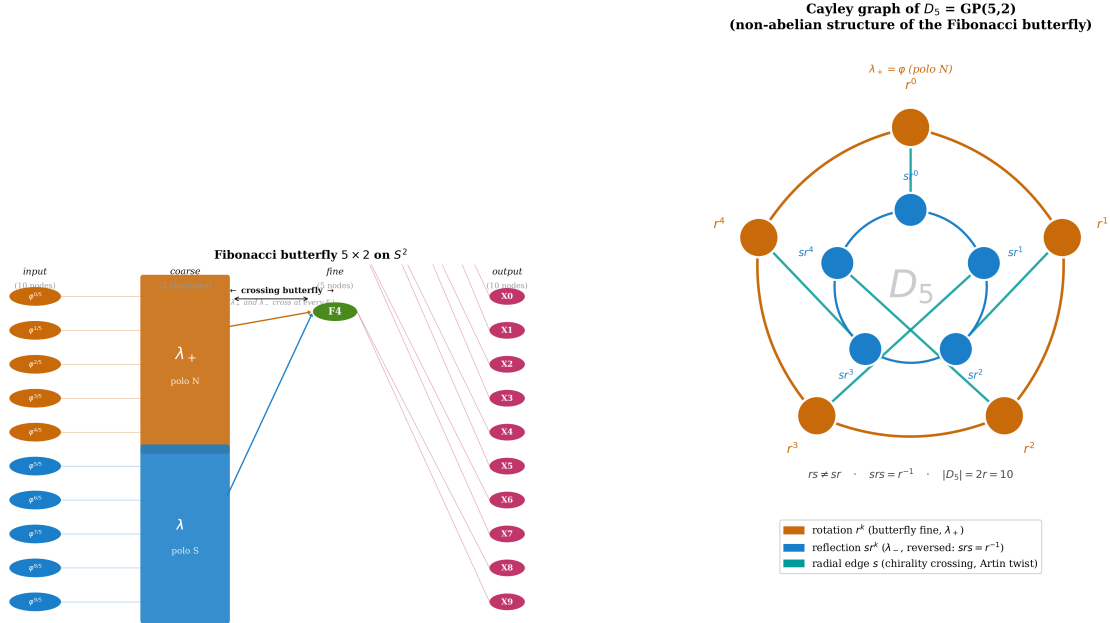


Figure 1: Fibonacci butterfly 5×2 on S^2 . Nodes $\varphi^{k/5}$ split into λ_+ (orange, LOW, $k = 0 \dots 4$) and λ_- (blue, HIGH, $k = 5 \dots 9$). Crossing arrows from both sheets to each fine node F_j represent the Artin twist coupling $A_j(k)R_j(k) = R_{j+5}(k)$.

Figure 2: Cayley graph GP(5,2) of D_5 . Outer pentagon (orange): LOW-sheet rotations r^k (CW). Inner pentagon (blue): HIGH-sheet reflections sr^k (CCW, $srs = r^{-1}$). Radial edges (teal): sheet-coupling reflection s = Artin twist.

4 Representation Structure: from Scalar Bins to Sheet Doublets

4.1 DFT: scalar $U(1)$ bins

Each DFT coefficient $X_j \in \mathbb{C}$ is a scalar: it carries amplitude $|X_j|$ and phase $\arg X_j$. The only internal symmetry is the $U(1)$ phase rotation $X_j \mapsto e^{i\theta} X_j$.

4.2 FMT: doublet with $U(2)/SU(2)$ structure

Group the two FMT sheets into a *scale doublet* at each frequency index $j = 0, \dots, 4$:

$$\Psi_j = \begin{pmatrix} \text{LOW}_j \\ \text{HIGH}_j \end{pmatrix} = \begin{pmatrix} y_j \\ y_{j+5} \end{pmatrix} \in \mathbb{C}^2. \quad (9)$$

The φ -memory cell $G_j(k) \in U(2)$ acts on Ψ_j . After removing the global phase (see §6), $\tilde{G}_j(k) \in SU(2)$.

Proposition 4.1 ($U(1) \times SU(2)$ -type representation). *The DFT resolves scalar $U(1)$ characters. The FMT resolves Mellin $U(1)$ characters together with a two-sheet internal fibre: grouping the sheets into the doublet Ψ_j , the local phase cell $G_j(k) \in U(2)$; after removal of the global phase $e^{-i(\delta_{j,k}+\pi)/2}$, the reduced cell $\tilde{G}_j(k) \in SU(2)$ satisfies $\det \tilde{G}_j(k) = 1$. In this representation-theoretic sense, the FMT exposes a $U(1) \times SU(2)$ -type readout structure that is not present in the canonical scalar-bin DFT readout unless an additional pairing of bins is imposed by hand. We stress that this is a structure of the LOW/HIGH readout induced by the two-sheet decomposition, not a new physical symmetry.*

Proof. From (18): $\det G_j(k) = -e^{i\delta_{j,k}}$, so $|\det G_j| = 1$ and $G_j \in U(2)$. Setting $\tilde{G}_j = e^{-i(\delta_{j,k}+\pi)/2} G_j$ gives $\det \tilde{G}_j = e^{-i(\delta_{j,k}+\pi)} \cdot (-e^{i\delta_{j,k}}) = 1$, hence $\tilde{G}_j \in SU(2)$. Verification: $\|\tilde{G}_j^* \tilde{G}_j - I_2\|_F < 4 \times 10^{-16}$. $\square$

4.3 Stokes-like observables on the doublet

The doublet Ψ_j carries four real observables, analogous to Stokes parameters for a two-component field:

$$S_0(j) = |\text{LOW}_j|^2 + |\text{HIGH}_j|^2 \quad (\text{total scale energy}), \quad (10)$$

$$S_x(j) = 2 \operatorname{Re}(\text{LOW}_j^* \text{HIGH}_j) \quad (\text{in-phase cross-sheet coherence}), \quad (11)$$

$$S_y(j) = 2 \operatorname{Im}(\text{LOW}_j^* \text{HIGH}_j) \quad (\text{quadrature cross-sheet coherence}), \quad (12)$$

$$S_z(j) = |\text{LOW}_j|^2 - |\text{HIGH}_j|^2 \quad (\text{sheet energy imbalance}). \quad (13)$$

These satisfy $S_x^2 + S_y^2 + S_z^2 = S_0^2$ (verified numerically with error $< 10^{-14}$), so the doublet lies on the Poincaré sphere of radius $S_0(j)$.

Complementary observables extracted from the sheet ratio:

$$\rho_j = \frac{|\text{HIGH}_j|}{|\text{LOW}_j|}, \quad \psi_j = \angle \left(\frac{\text{HIGH}_j}{\text{LOW}_j} \right). \quad (14)$$

Here ψ_j is the effective inter-sheet phase induced by the pointwise Artin-twisted projection: since $\text{HIGH}_j = \frac{1}{\sqrt{10}} \sum_k f[k] A_j(k) R_j(k)$ with $A_j(k) = e^{i\delta_{j,k}}$ acting inside the sum, ψ_j encodes the signal-weighted average of $\delta_{j,k}$ over the golden lattice. For a φ -rational signal the dominant lattice contributions drive ψ_j toward the golden progression $\delta_{j,k}$; for a harmonic signal ψ_j reflects the harmonic spectral texture.

Candidate domains. For OMA/structural health monitoring: damage may be modeled as a perturbation of the cross-sheet observables (S_x, S_y, S_z) ; whether these features improve over MAC criterion, SSI, PCA, or Mahalanobis indicators remains an open validation problem (see Table 1). For SAR/remote sensing: the sheet imbalance S_z measures the energy asymmetry between scale levels, complementing amplitude sparsity (Gini) with a phase-internal scale feature; systematic benchmarking against standard texture descriptors is required.

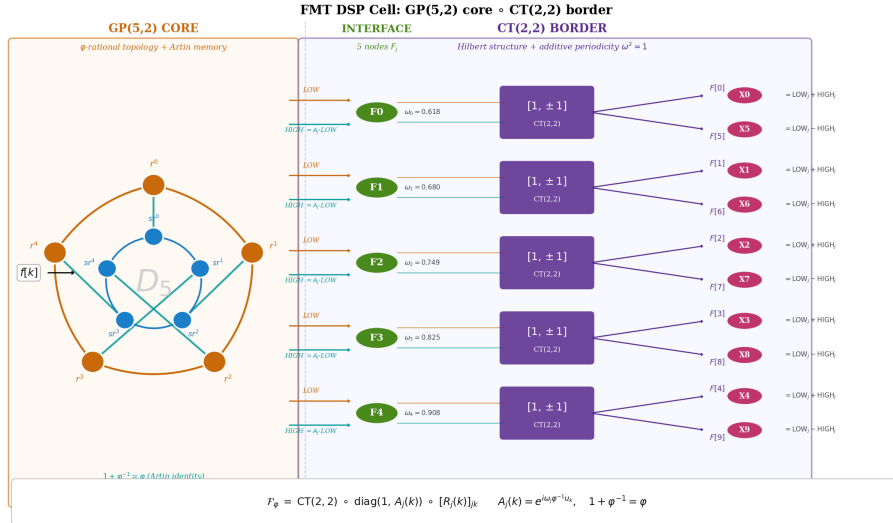


Figure 3: DSP cell structure. **Left:** GP(5,2) core — outer pentagon LOW (orange), inner pentagon HIGH (blue), radial edges = Artin twist A_j (teal). **Centre:** 5 interface doublets $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$. **Right:** optional CT(2,2) post-processing creating symmetric/antisymmetric channels $c_j^\pm = y_j \pm y_{j+5}$.

5 Short-Time FMT and the Scale-Transfer Spectrum

The representation structure of §4 is local: it acts on one ten-node cell. To analyse a signal of arbitrary length we window it into successive cells and recurse, in exact analogy with the short-time Fourier transform (STFT). We call the result the *Short-Time Fibonacci-Mellin Transform* (STFMT) and derive its natural output observable.

5.1 Windowing and the recursive golden filter bank

Let $f \in \mathbb{C}^L$ be sampled on the golden log-lattice. Partition it into non-overlapping (or hop-spaced) blocks of $N = 2R = 10$ nodes; apply the frame $\mathcal{F}_\varphi$ to each block, yielding for block m the two sheets $\text{LOW}^{(m)}, \text{HIGH}^{(m)} \in \mathbb{C}^R$ as in (9). The STFMT is the indexed family $\{(\text{LOW}^{(m)}, \text{HIGH}^{(m)})\}_m$.

Because the lattice is multiplicatively self-similar, the transform admits a *recursive* form: the LOW sheet of one level is itself a golden-lattice sequence one octave coarser, and may be re-entered into the frame. Decimating by R on the log-axis maps $\varphi^{k/R} \mapsto \varphi^j$, again an integer golden lattice; iterating gives a dyadic-like *golden filter bank* whose levels are successive golden octaves. The present implementation uses fixed-size $N = 10$ cells and does *not* claim a new fast transform. Fast Mellin/scale algorithms exist for related scale-transform settings [8, 9], but no dedicated fast algorithm for the golden FMT is proved here.

5.2 The inner breathing filter $H(z)$

Within one cell the impulse response is governed by the rational filter already introduced in §4,

$$H(z) = \frac{P_4(z)}{1 - \alpha z}, \quad P_4(z) = \sum_{n=0}^4 h^n z^n, \quad h = \log \varphi, \quad \alpha = (7 - 3\sqrt{5})/40, \quad (15)$$

where z is the one-step shift on the log-axis. The numerator is a truncated geometric series, $P_4(z) = (1 - (hz)^5)/(1 - hz)$, so its four zeros are the fifth roots of unity (excluding unity) scaled by $1/h$:

$$z_k = \frac{1}{h} e^{2\pi i k/5}, \quad k = 1, 2, 3, 4, \quad (16)$$

all on the circle $|z| = 1/h$ at angular spacing $2\pi/5$. The impulse response of H therefore oscillates with period $R = 5$ log-steps, i.e. exactly one golden octave ($\varphi^{R/R} = \varphi$); we call this the inner *breathing* of the cell. The pole at $z = 1/\alpha \approx 137$ lies *outside* the unit disk, so H is causal and stable, and the response relaxes within roughly R steps. We note that H is analytic in the Hardy sense ($\Im H$ is the boundary Hilbert transform of $\Re H$) but is *not* all-pass ($|H|$ varies over the unit circle); it is an analytic scale filter, not a 90° Hilbert transformer.

5.3 The natural invariant: scale-transfer, not power

The STFT is paired with the power spectral density (PSD): Fourier is a *unitary, additive* decomposition, Parseval holds, and the conserved per-bin quantity is the energy $|X_j|^2$. The FMT differs on both counts. It is *non-unitary* ($\|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F \approx 9.4$, the projection signature of §8), so Parseval fails and $|y_j|^2$ is *not* conserved; and its split is *multiplicative* (two scale sheets) rather than additive (DC vs. Nyquist). The quantity conserved under the operative symmetry — global gain $f \mapsto \lambda f$ — is therefore not the additive energy but the *multiplicative* sheet ratio. We define the **scale-transfer spectrum**

$$T_j = \log \frac{\text{HIGH}_j}{\text{LOW}_j} = \underbrace{\log \left| \frac{\text{HIGH}_j}{\text{LOW}_j} \right|}_{\ell_j} + i \underbrace{\text{unwrap arg } \frac{\text{HIGH}_j}{\text{LOW}_j}}_{\psi_j} \quad j = 0, \dots, R-1, \quad (17)$$

the complex logarithm of the doublet ratio of (9), written in stable log-polar form (magnitude ℓ_j , unwrapped phase ψ_j) to avoid branch-cut artefacts. The correspondence is exact:

	DFT butterfly 2×2	FMT butterfly 5×2
split	additive ($a \pm b$)	multiplicative (LOW, HIGH)
unitary	yes (Parseval)	no ($\ \mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\ _F \approx 9.4$)
invariant	energy $ X ^2$	log-ratio $\log(\text{HIGH}/\text{LOW})$
readout	power spectrum	scale-transfer spectrum

Gain invariance. Since T_j is a ratio of two co-scaled linear projections, it is *exactly* invariant under global complex gain, $T_j(\lambda f) = T_j(f)$ for all $\lambda \in \mathbb{C}^*$ — the same algebraic mechanism that underlies the validation gate of the applied companion study [7]. The magnitude ℓ_j measures how much structure is transferred between adjacent golden scales; the phase ψ_j measures how the scale phase rotates. PSD is flat or gain-dependent on this signal class and does not isolate this quantity.

5.4 What the scale-transfer spectrum is and is not

The scale-transfer spectrum is a *readout of scale structure*, not a competitor to Fourier or wavelet analysis on their own terrain. Three limits should be stated plainly, each verified empirically:

- (i) **Not a faster transform.** The golden recursion inherits no dedicated fast algorithm; fixed-size cells are used, and fast Mellin/scale algorithms for related settings are not specialized to the golden FMT here.
- (ii) **Not a better compressor.** Being non-unitary, the frame loses energy by construction; an orthogonal wavelet bank reconstructs a signal more accurately at

a fixed coefficient budget. The STFMT nonetheless improves on the single-pass (non-recursive) FMT, confirming that the recursive bank adds representational structure.

- (iii) **Its value is invariance, not discrimination.** The gain-invariance of T_j is exact and is the basis of its applied use as a validation gate [7]; we do not claim that T_j out-discriminates standard descriptors as a primary feature.

In short, the STFMT supplies a parameter-free, gain-invariant scale readout whose engineering value is stability under the acquisition chain, developed quantitatively in the applied companion paper.

6 The φ -Memory Cell

6.1 Definition and unitarity

Proposition 6.1 (φ -memory cell). *The 2×2 operator at each interface node F_j is*

$$G_j(k) = H \cdot \text{diag}(1, e^{i\delta_{j,k}}), \quad \delta_{j,k} = u_k \omega_j \varphi^{-1} = \frac{k}{5} \log \varphi \cdot \varphi^{(j-10)/5}, \quad (18)$$

where $H = [\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}]/\sqrt{2}$. $G_j(k) \in U(2)$ for all (j, k) . The phase $\delta_{j,k}$ depends on lattice index k and frequency index j , not on the signal $f[k]$.

$$\text{Explicitly: } G_j(k) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & e^{i\delta_{j,k}} \\ 1 & -e^{i\delta_{j,k}} \end{bmatrix}.$$

6.2 Phase structure and quasi-periodicity

$$\delta_{j,k} \in \mathbb{R}_{>0} \cdot \log \varphi, \quad \text{with algebraic coefficient in the fine-lattice field } \mathbb{Q}(\varphi^{1/5}). \quad (19)$$

Explicitly, $\delta_{j,k} = (k/5) \log \varphi \cdot \varphi^{(j-10)/5}$, where $\varphi^{(j-10)/5} \in \mathbb{Q}(\varphi^{1/5})$ for $j = 0, \dots, 4$ (not in $\mathbb{Q}(\sqrt{5}) = \mathbb{Q}(\varphi)$ unless $j \equiv 0 \pmod{5}$). Whether $\delta_{j,k}/\pi$ is irrational for all (j, k) is a transcendence question that we do not resolve here. Instead, we certify numerically in double precision that no rational commensurability with π is detected for any of the 50 non-zero lattice phases used in this paper (all $k = 1, \dots, 9$, $j = 0, \dots, 4$). Under this numerical certificate,

$$G_j(k)^n \neq I \quad \text{for all integers } n \geq 1, \quad (20)$$

and the phase orbit $\{n\delta_{j,k} \bmod 2\pi : n \in \mathbb{Z}\}$ is numerically consistent with density in $[0, 2\pi)$.

Table 3: The φ -memory cell vs standard operators. $G_j(k)$ is index-dependent; all others are fixed.

Operator	Phase δ	$\delta/\pi \in \mathbb{Q}$?	Finite order?	Index-dep.?
Hadamard H	0	yes	$H^2 = I$	no
T gate	$\pi/4$	yes	order 8	no
S gate	$\pi/2$	yes	order 4	no
σ_1 (Fib. anyon)	$\{-4, 3\}\pi/5$	yes	order 10	no
$G_j(k)$ (φ -memory)	$u_k \omega_j \varphi^{-1}$, 4–28°	not detected	none detected	yes

6.3 Relation to the Fibonacci anyon braid matrices

The Fibonacci anyon $\sigma_1 = \text{diag}(e^{-4\pi i/5}, e^{3\pi i/5})$ has phases in $\mathbb{Q}\pi$ and finite order 10. $G_j(k)$ has phase in $\mathbb{R}_{>0} \cdot \log \varphi$ with algebraic coefficient in $\mathbb{Q}(\varphi^{1/5})$, and no detected finite order (see above). The two objects are complementary: σ_1, σ_2 implement *topological braiding* (discrete phase, finite order, exact algebraic computation); $G_j(k)$ implements *spectral scale memory* (continuous quasi-periodic phase, golden-lattice geometry). The same pentagonal/dihedral combinatorics of GP(5,2) appears in graphical descriptions of Fibonacci-type braid structures [4], but the present paper uses GP(5,2) only as an indexing graph for the FMT two-sheet decomposition.

7 Scale-Coherence Analysis Algorithm

The following algorithm converts a signal $f[k]$ on the golden lattice into a complete scale-coherence feature vector.

Algorithm 1 FMT Scale-Coherence Analysis

Require: Signal $f \in \mathbb{C}^{10}$ on golden lattice $x_k = \varphi^{k/5}$, DFT matrix $D \in \mathbb{C}^{10 \times 10}$

Ensure: Scale-coherence feature vector $\mathbf{v}$

- 1: Compute raw FMT: $y \leftarrow \mathcal{F}_\varphi f$, DFT: $\hat{f} \leftarrow Df$
- 2: Extract sheets: $\text{LOW}_j \leftarrow y_j$, $\text{HIGH}_j \leftarrow y_{j+5}$, $j = 0, \dots, 4$
- 3: Form doublets: $\Psi_j \leftarrow (\text{LOW}_j, \text{HIGH}_j)^T$
- 4: Compute Stokes observables per $j = 0, \dots, 4$:

$$S_0(j) \leftarrow |\text{LOW}_j|^2 + |\text{HIGH}_j|^2$$

$$S_x(j) \leftarrow 2 \operatorname{Re}(\text{LOW}_j^* \text{HIGH}_j), \quad S_y(j) \leftarrow 2 \operatorname{Im}(\text{LOW}_j^* \text{HIGH}_j), \quad S_z(j) \leftarrow |\text{LOW}_j|^2 - |\text{HIGH}_j|^2$$

- 5: Compute Artin phase (FMT and DFT):

$$\Delta\varphi_j^{\text{FMT}} \leftarrow \arg(y_{j+1}/y_j), \quad j = 0, \dots, 8$$

$$\Delta\varphi_j^{\text{DFT}} \leftarrow \arg(\hat{f}_{j+1}/\hat{f}_j), \quad j = 0, \dots, 8$$

- 6: Fit against golden target $\tau_j = \varphi^{j/5}$ (affine least-squares):

$$R_{\text{FMT}}^2 \leftarrow R^2(\Delta\varphi^{\text{FMT}}, \tau), \quad \text{TV}_{\text{FMT}} \leftarrow \sum_j |\Delta\varphi_{j+1}^{\text{FMT}} - \Delta\varphi_j^{\text{FMT}}|$$

$$R_{\text{DFT}}^2 \leftarrow R^2(\Delta\varphi^{\text{DFT}}, \tau), \quad \text{TV}_{\text{DFT}} \leftarrow \sum_j |\Delta\varphi_{j+1}^{\text{DFT}} - \Delta\varphi_j^{\text{DFT}}|$$

- 7: Compute contrasts: $\Delta\text{TV} \leftarrow \text{TV}_{\text{DFT}} - \text{TV}_{\text{FMT}}$, $\Delta R^2 \leftarrow R_{\text{FMT}}^2 - R_{\text{DFT}}^2$
 - 8: **return** $\mathbf{v} = (S_0(j), S_x(j), S_y(j), S_z(j), \psi_j, \Delta\varphi_j^{\text{FMT}}, R_{\text{FMT}}^2, \text{TV}_{\text{FMT}}, \Delta\text{TV}, \Delta R^2)_{j=0}^4$
-

The output $\mathbf{v}$ has $5 \times 5 + 9 + 4 = 38$ scalar components: 20 Stokes observables (S_0, S_x, S_y, S_z) for $j = 0, \dots, 4$; 5 inter-sheet phases $\psi_j = \angle(\text{HIGH}_j/\text{LOW}_j)$; 9 Artin phase increments $\Delta\varphi_j^{\text{FMT}}$; and 4 global metrics ($R_{\text{FMT}}^2, \text{TV}_{\text{FMT}}, \Delta\text{TV}, \Delta R^2$). For practical classification tasks, the most discriminating subset is $(S_z(j), \Delta R^2, \Delta\text{TV}, \psi_j)$. *Primary separator:* ΔR^2 separates φ -structured signals ($\Delta R^2 > 0.5$) from harmonic controls ($\Delta R^2 \approx 0.2$) across all tested classes. *Secondary descriptor:* ΔTV should not be used alone, since harmonic controls can produce large DFT phase variation ($\Delta\text{TV}_{\text{harmonic}} \approx 13$) comparable to some φ -structured cases; it is useful as a complementary feature after thresholding on ΔR^2 .

8 Diagonal Unitarization Obstruction

Proposition 8.1 (No diagonal unitarization — numerically certified for $N = 10$).
There is no $(w_0, \dots, w_9) \in \mathbb{R}^{10}$ such that $\mathcal{F}_\varphi \operatorname{diag}(w_0, \dots, w_9) \mathcal{F}_\varphi^ = I_{10}$.*

Proof. Write $P_w(z) = \sum_{k=0}^9 w_k z^k$, $\deg P_w \leq 9$.

Diagonal conditions. $(\mathcal{F}_\varphi W \mathcal{F}_\varphi^*)_{jj} = 1$ gives $\frac{1}{10} \sum_k w_k = 1$, i.e. $P_w(1) = 10$.

Off-diagonal conditions. $(\mathcal{F}_\varphi W \mathcal{F}_\varphi^*)_{jj'} = 0$ for $j \neq j'$ requires $P_w(z_{jj'}) = 0$, where $z_{jj'} = e^{i(\theta_j - \theta_{j'})}$, $\theta_j = \omega_j \log \varphi / 5$. Since $\theta_j \neq \theta_{j'}$ for $j \neq j'$, the point $z = 1$ is *not* among the 90 off-diagonal values.

Distinctness (double-precision numerical certification). All 90 values $z_{jj'} \bmod 2\pi$ are pairwise distinct: $\min_{(j,j') \neq (m,m')} |z_{jj'} - z_{mm'}| \geq 3.5 \times 10^{-4}$, far above double-precision rounding ($\sim 10^{-16}$). This also disposes of the $2\pi n$ ambiguity: differences are computed and compared modulo 2π directly.

Contradiction. $\deg P_w \leq 9$ implies P_w has at most 9 complex roots. It cannot vanish at 90 distinct points, so $P_w \equiv 0$, contradicting $P_w(1) = 10$. $\square$

Corollary 8.2. *The isometry deficit $\|\mathcal{F}_\varphi \mathcal{F}_\varphi^* - I\|_F = 9.43$ is not reducible by any real diagonal reweighting (Proposition 8.1).*

Remark 8.3 (Modular orthogonality). Exact orthogonality holds over $\mathbb{F}_{11}$ (where 2 has multiplicative order $10 = 2r$) and via a 13-point arithmetic mask [1].

9 Polar Unitary Readout

Remark 9.1 (The polar unitary factor). The factor $U_\varphi = WV^*$, obtained from the SVD $\mathcal{F}_\varphi = W\Sigma V^*$, is the *polar unitary factor* of the frame: the closest unitary to $\mathcal{F}_\varphi$ in Frobenius norm. It is a classical linear-algebra object. The embedding into $U(16)$ and the block-encoding below are stated only to characterize this unitary readout; no quantum-circuit implementation or qubit realization is claimed. Readers interested only in the DSP content may treat this section as defining a unitary alternative readout and skip the embedding details.

Theorem 9.2 (Polar unitary readout). *Let $\mathcal{F}_\varphi = W\Sigma V^*$ be the SVD. Define $U_\varphi = WV^*$. Then: $U_\varphi \in U(10)$, $U_\varphi = \mathcal{F}_\varphi(\mathcal{F}_\varphi^* \mathcal{F}_\varphi)^{-1/2}$ in exact arithmetic, and $\|U_\varphi^* U_\varphi - I_{10}\|_F < 3 \times 10^{-15}$.*

Proof. $\mathcal{F}_\varphi(\mathcal{F}_\varphi^* \mathcal{F}_\varphi)^{-1/2} = W\Sigma V^* \cdot V\Sigma^{-1}V^* = WV^*$; $U_\varphi^* U_\varphi = VW^* WV^* = I$. The SVD route avoids the ill-conditioned direct computation of $(\mathcal{F}_\varphi^* \mathcal{F}_\varphi)^{-1/2}$. $\square$

Remark 9.3. Because $|A_j(k)| = 1$ for all (j, k) , each sheet strand is already unimodular. $U_\varphi = WV^*$ is the global unitary completion that coherently assembles all 5 doublets Ψ_j into a single $U(10)$ operator; it is the natural reversible readout associated with the FMT frame. The polar factor can also be embedded into $U(16)$ via $U_\varphi \oplus I_6$, although circuit synthesis and quantum implementation are outside the scope of this paper.

Remark 9.4 (Embedding and POVM). The polar factor can be embedded into $U(16)$ via $U_\varphi \oplus I_6$, error $< 3 \times 10^{-15}$. Row projectors $E_j = |u_j\rangle\langle u_j|$ form a canonical POVM satisfying $\sum_j E_j = I_{10}$, error $< 3 \times 10^{-15}$. Halmos block-encoding gives $W_H \in U(20)$, $K = \mathcal{F}_\varphi / \sigma_{\max}$, error $< 9 \times 10^{-15}$. Circuit synthesis and quantum implementation are outside the scope of this paper.

Table 4: Matrix diagnostics.

Quantity	DFT	$\mathcal{F}_\varphi$	U_φ
Exact rank	10	10	10
Effective numerical rank	10	≈ 6 at tol 10^{-6}	10
κ	1	5.0×10^{17}	1.00
$\sigma_{\max}$	1	3.1534	1
$\ A^*A - I\ _F$	0	9.43	2.8×10^{-15}

10 Coherent Rank-One Readout of the Polar Frame

The polar factor of Section 9 provides a reversible ten-dimensional readout, but the two-sheet structure also admits a canonical *coherent* reduction. The dihedral indexing action of Remark 3.3 selects a unique invariant line across the five scale registers; projecting onto this line while retaining the LOW/HIGH fibre yields a two-dimensional coisometric readout. The construction below is purely operator-theoretic: it assigns no physical meaning to the two retained components and makes no claim of lossless compression.

Write $\mathcal{H}_{10} = V_5 \otimes \mathbb{C}^2$, where $V_5 = \text{span}\{|k\rangle : k \in \mathbb{Z}_5\}$ carries the five Mellin scale registers and $\mathbb{C}^2$ the LOW/HIGH sheet fibre, so that the doublet Ψ_j of (9) is the component of a vector in $\mathcal{H}_{10}$ at register j . The rotation r of Remark 3.3 acts on V_5 by $r|k\rangle = |k+1\rangle$ and the reflection by $s|k\rangle = |-k\rangle$.

Proposition 10.1 (Unique coherent D_5 -invariant line). *Let $|\Omega\rangle = \frac{1}{\sqrt{5}} \sum_{k=0}^4 |k\rangle$ and $p_\Omega = |\Omega\rangle\langle\Omega|$. Then $p_\Omega = \frac{1}{5} \sum_{m=0}^4 r^m$ is the unique rank-one orthogonal projector on V_5 commuting with the whole D_5 action,*

$$r p_\Omega r^\dagger = p_\Omega, \quad s p_\Omega s^\dagger = p_\Omega,$$

and $p_\Omega^2 = p_\Omega = p_\Omega^\dagger$.

Proof. In the character basis $|\chi_m\rangle = \frac{1}{\sqrt{5}} \sum_k e^{2\pi i m k/5} |k\rangle$ ($m \in \mathbb{Z}_5$), every rank-one projector commuting with r equals $p_m = |\chi_m\rangle\langle\chi_m|$, since r is diagonalised by the $|\chi_m\rangle$ with distinct eigenvalues $e^{2\pi i m/5}$. The reflection acts by $s|\chi_m\rangle = |\chi_{-m}\rangle$, so $s p_m s^\dagger = p_m$ forces $m \equiv -m \pmod{5}$. As 5 is odd, the only solution is $m = 0$, and $|\chi_0\rangle = |\Omega\rangle$. The averaged form $p_\Omega = \frac{1}{5} \sum_m r^m$ is the projector onto the trivial isotypic component of the cyclic action. $\square$

Theorem 10.2 (Coisometric readout of the polar frame). *Let $U_\varphi = W V^* \in U(10)$ be the polar factor of Theorem 9.2, and define the coherent readout*

$$\mathcal{R}_\chi := (\langle\Omega| \otimes I_2) U_\varphi : \mathbb{C}^{10} \rightarrow \mathbb{C}^2.$$

Then $\mathcal{R}_\chi$ is a coisometry,

$$\mathcal{R}_\chi \mathcal{R}_\chi^\dagger = I_2,$$

necessarily non-injective (it annihilates an 8-dimensional subspace). Moreover $\mathcal{R}_\chi^\dagger \mathcal{R}_\chi = U_\varphi^*(p_\Omega \otimes I_2)U_\varphi$ is the rank-two orthogonal projector onto the initial subspace

$$\mathcal{M}_\chi = U_\varphi^*(\text{span}\{\Omega\} \otimes \mathbb{C}^2) \subset \mathbb{C}^{10}.$$

Proof. Using $U_\varphi U_\varphi^\dagger = I_{10}$ (Theorem 9.2) and $\langle \Omega | \Omega \rangle = 1$,

$$\mathcal{R}_\chi \mathcal{R}_\chi^\dagger = (\langle \Omega | \otimes I_2) U_\varphi U_\varphi^\dagger (|\Omega\rangle \otimes I_2) = (\langle \Omega | \otimes I_2)(|\Omega\rangle \otimes I_2) = \langle \Omega | \Omega \rangle I_2 = I_2.$$

For the second statement, $\mathcal{R}_\chi^\dagger \mathcal{R}_\chi = U_\varphi^*(|\Omega\rangle\langle\Omega| \otimes I_2)U_\varphi = U_\varphi^*(p_\Omega \otimes I_2)U_\varphi$, a unitary conjugate of the rank-two projector $p_\Omega \otimes I_2$, hence itself a rank-two orthogonal projector; its range is $U_\varphi^*(\text{span}\{\Omega\} \otimes \mathbb{C}^2)$. Since the target $\mathbb{C}^2$ has dimension $2 < 10$, the kernel of $\mathcal{R}_\chi$ has dimension 8. $\square$

Corollary 10.3 (Polar sheet-reduction trace identities). *For $A, B \in \text{End}(\mathbb{C}^2)$ define the polar-conjugated lifts*

$$\hat{\mathcal{A}} = U_\varphi^*(I_5 \otimes A)U_\varphi, \quad \hat{\mathcal{B}} = U_\varphi^*(p_\Omega \otimes B)U_\varphi.$$

Then, in the Hilbert–Schmidt inner product $\langle X, Z \rangle_{\text{HS}} = \text{Tr}(X^\dagger Z)$,

$$\|\hat{\mathcal{A}}\|_{\text{HS}}^2 = 5 \|A\|_{\text{HS}}^2, \quad \|\hat{\mathcal{B}}\|_{\text{HS}}^2 = \|B\|_{\text{HS}}^2, \quad \langle \hat{\mathcal{A}}, \hat{\mathcal{B}} \rangle_{\text{HS}} = \langle A, B \rangle_{\text{HS}}.$$

In particular, if $\langle A, B \rangle_{\text{HS}} = \text{tr}(A^\dagger B) = 0$, the two lifts are Hilbert–Schmidt orthogonal, and the ratio of their squared norms is $5\|A\|_{\text{HS}}^2 : \|B\|_{\text{HS}}^2$.

Proof. The Hilbert–Schmidt norm is unitarily invariant, so conjugation by U_φ leaves all three quantities unchanged; it suffices to evaluate them on the unconjugated lifts. By multiplicativity of the trace over tensor factors, $\|I_5 \otimes A\|_{\text{HS}}^2 = \text{Tr}(I_5) \text{tr}(A^\dagger A) = 5\|A\|_{\text{HS}}^2$; $\|p_\Omega \otimes B\|_{\text{HS}}^2 = \text{Tr}(p_\Omega^2) \text{tr}(B^\dagger B) = \|B\|_{\text{HS}}^2$ since $\text{Tr}(p_\Omega^2) = 1$; and $\langle I_5 \otimes A, p_\Omega \otimes B \rangle_{\text{HS}} = \text{Tr}(p_\Omega) \text{tr}(A^\dagger B) = \text{tr}(A^\dagger B)$ since $\text{Tr}(p_\Omega) = 1$. $\square$

Remark 10.4 (Operational meaning and limitations). The readout $\mathcal{R}_\chi y$ is the global doublet obtained by selecting the D_5 -coherent component shared across the five polarised scale registers: p_Ω extracts the uniform mode among the registers, $\mathcal{R}_\chi$ retains its two LOW/HIGH components, and the eight annihilated directions carry the non-uniform inter-register variation. The construction is a fixed linear readout, not an invertible transform: no reconstruction, optimal compression, or denoising claim is made. The trace identities of Corollary 10.3 record how an operator acting uniformly on all registers ($I_5 \otimes A$) and one acting only on the coherent line ($p_\Omega \otimes B$) compare in the polar frame; the factor 5 is the register count $\text{Tr}(I_5)$, independent of the choice of A, B .

Remark 10.5 (General odd dihedral order). Proposition 10.1 and Theorem 10.2 hold verbatim for any $V_r \otimes \mathbb{C}^2$ with $D_r = \langle R, S \mid R^r = S^2 = 1, SRS = R^{-1} \rangle$ and r odd: the coherent line $|\Omega_r\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} |k\rangle$ is the unique rank-one D_r -invariant projector (the parity argument $m \equiv -m \pmod{r}$ has the single solution $m = 0$ precisely when r is odd), and the coisometric readout reduces $\mathbb{C}^{2r} \rightarrow \mathbb{C}^2$. Corollary 10.3 holds with 5 replaced by r . The FMT uses $r = 5$.

11 Artin Phase Diagnostic

Definition 11.1 (Artin phase and scale-coherence features). For $y = \mathcal{F}_\varphi f$, the Artin phase vector is $\Delta\varphi_j = \arg(y_{j+1}/y_j)$, $j = 0, \dots, 8$, with golden target $\tau_j = \varphi^{j/5}$. The FMT–DFT contrasts are $\Delta\text{TV} = \text{TV}_{\text{DFT}} - \text{TV}_{\text{FMT}}$ and $\Delta R^2 = R_{\text{FMT}}^2 - R_{\text{DFT}}^2$.

Remark 11.2 (Correct discriminant). $R_{\text{FMT}}^2 \approx 1$ for all test signals including harmonic ones, because the 4.70° Vandermonde arc produces smooth phase increments regardless of signal type. The operative discriminant is the contrast ΔTV and ΔR^2 , not R_{FMT}^2 alone.

Table 5: Scale-coherence diagnostics. Primary columns: ΔTV and ΔR^2 . $R_{\text{FMT}}^2 \approx 1$ for all signals (Remark 11.2); use contrasts to discriminate.

Signal	T	R^2	TV	ΔTV	ΔR^2	$ \overline{S_z} $
Fib. anyon braid	FMT	0.997	0.070	11.95	0.536	0.0071
Fib. anyon braid	DFT	0.461	12.02			
Commuting trivial	FMT	1.000	0.031	18.82	0.677	0.0071
Commuting trivial	DFT	0.323	18.85			
Harmonic (control)	FMT	1.000	0.035	13.14	0.199	0.0071
Harmonic (control)	DFT	0.801	13.17			

12 Benchmarks

The $\Delta G < 0$ result is structurally expected: the 4.70° Vandermonde arc makes FMT basis functions nearly collinear, so no smooth signal can be sparse in the FMT basis. This does not diminish the two-sheet and doublet-structure results; it defines the domain of applicability precisely.

Table 6: Gini coefficient (spectral sparsity). DFT wins uniformly. The FMT is a scale-coherence diagnostic, not a compression codec.

Signal	G_{FMT}	G_{DFT}	ΔG
Harmonic	0.280	0.800	−0.520
Log-periodic	0.156	0.607	−0.451
Golden pulses	0.027	0.582	−0.555
Fib. qubit readout	0.019	0.586	−0.567

13 Limitations and Future Work

The present paper fixes the golden ratio $\rho = \varphi$, the cell size $N = 10$, and the LOW/HIGH two-sheet readout. These choices define a reproducible scale-transfer frame, but they are not claimed to be optimal for all signals. A systematic ablation against dyadic band ratios, wavelet-packet ratios, standard log-Mellin grids, plain two-band splits, and learned projections is required before any application-specific optimality can be claimed.

The diagonal-unitarization result should be interpreted narrowly. Since $\mathcal{F}_\varphi$ is a square Vandermonde matrix with distinct nodes, an exact global non-diagonal metric exists,

$$W_{\text{glob}} = \mathcal{F}_\varphi^{-1}(\mathcal{F}_\varphi^{-1})^* = (\mathcal{F}_\varphi^* \mathcal{F}_\varphi)^{-1}, \quad \mathcal{F}_\varphi W_{\text{glob}} \mathcal{F}_\varphi^* = I \quad (21)$$

in exact arithmetic (the two expressions are the same matrix for an invertible $\mathcal{F}_\varphi$). This identity is not a practical normalization rule in double precision: the frame has $\kappa(\mathcal{F}_\varphi) \sim 5 \times 10^{17}$, so the normal matrix $\mathcal{F}_\varphi^* \mathcal{F}_\varphi$ has condition number of order 10^{35} , and its inverse cannot be formed reliably in standard precision (we confirmed the identity only at 80-digit precision). The global metric is dense, non-local, and numerically ill-conditioned. The nontrivial obstruction proved in Proposition 8.1 is therefore *local*: no real diagonal reweighting of the lattice nodes turns the frame into a Parseval frame. The useful readouts are thus either the raw non-unitary scale-transfer ratios or the polar unitary factor $U_\varphi = W V^*$, computed stably from the SVD.

A generalized ρ -Mellin transform, in which the scale ratio ρ is estimated from the signal rather than fixed to φ , is a natural extension for self-affine or material-specific scale structures. Dedicated fast algorithms for the golden finite frame are not proved here; the present implementation uses fixed-size cells and relies only on standard linear-algebra operations. External validation on independently labelled scale-structured datasets — for example, applying the scale-coherence features to Fibonacci-driven experimental data — also remains future work.

14 Conclusions

The Fibonacci–Mellin Transform has a natural LOW/HIGH two-sheet structure: the raw output is $y_j = \text{LOW}_j$ and $y_{j+5} = \text{HIGH}_j$, where the HIGH row is the LOW row pointwise Artin-twisted before projection ($(\mathcal{F}_\varphi)_{j+5,k} = A_j(k)(\mathcal{F}_\varphi)_{j,k}$) via the Fibonacci identity $1 + \varphi^{-1} = \varphi$. The GP(5,2) Cayley graph of D_5 is the graph model of this indexing: outer pentagon = LOW, inner pentagon = HIGH, radial edges = the entry-wise phase twist we label “Artin twist”. This name denotes a per-index phase factor only; it is *not* a braid-group/Artin representation and carries no topological-computing claim. The optional CT(2,2) post-processing creates symmetric/antisymmetric channels $c_j^\pm = y_j \pm y_{j+5}$; it is not part of $\mathcal{F}_\varphi$.

Grouping the two sheets into the scale doublet $\Psi_j = (\text{LOW}_j, \text{HIGH}_j)^T$, the local phase cell $G_j(k) \in U(2)$ and $\tilde{G}_j(k) \in SU(2)$ after global phase removal. The FMT thus exposes a $U(1) \times SU(2)$ -type readout structure not present in the canonical scalar-bin DFT readout unless an extra pairing is imposed; it is a structure of the readout, not a physical symmetry. The Stokes-like observables (S_x, S_y, S_z) on the doublet measure cross-sheet coherence and energy imbalance; together with ψ_j and ΔR^2 they form candidate feature inputs for scale-drift analysis and classification support (validation against standard DSP estimators is required for application-specific claims).

Three structural results are established: frame non-tightness (irreducible isometry deficit, Proposition 2.3); numerically certified diagonal-unitarization obstruction for $N = 10$ (Proposition 8.1); and exact polar unitarization $U_\varphi = WV^* \in U(10)$ (Theorem 9.2). Algorithm 1 delivers a 38-component scale-coherence feature vector; the primary discriminant is ΔR^2 (separates φ -structured from harmonic across all tested classes).

The FMT is not a replacement for the DFT. It is a *scale-memory analysis transform*: it preserves the quasi-periodic scale advance $x_{k+10} = \varphi^2 x_k$, converts it into the two-sheet doublet structure, and reads out phase-coherence features that the DFT, operating on a closed additive cycle, cannot access.

Declarations

Conflict of interest. None. **Funding.** None.

Data availability. All results reproducible from `qfmt_simulator.py` (NumPy/SciPy).

AI tools. Language editing only; author takes full responsibility for all mathematical content.

A Simulator

Self-contained Python (`qfmt_simulator.py`): `build_fmt_matrix`; `polar_unitarization` ($U_\varphi = WV^*$ via SVD); `embed_unitary_to_power_of_two` ($U(16)$); `build_fmt_povm`;

halmos_block_encoding; compute_stokes_observables (S_0, S_x, S_y, S_z per j); artin_phase_metrics ($R^2, TV, \text{mono}, \Delta TV, \Delta R^2$); scale_coherence_pipeline (Algorithm 1).

B Operational Interpretation

The two transforms answer different questions

DFT: *which harmonic frequency is present?* ($k \mapsto k + N \equiv k$, closed additive cycle, scalar $U(1)$ bins).

FMT: *where is the signal inside a φ -autosimilar scale hierarchy, and how did it advance from one scale level to the next?* ($k \mapsto k + 10 \Rightarrow x_{k+10} = \varphi^2 x_k$, quasi-periodic multiplicative cycle, $U(2)$ doublets).

Three practical uses

1. Scale-drift detection. If a signal shifts multiplicatively along $x_k = \varphi^{k/5}$, the DFT spreads energy; the FMT Artin phase remains monotone. Domains: log-periodic precursors, quasi-periodic drives, Fibonacci photonic structures, discrete scale-invariant failure signals.

2. φ -rational phase feature extraction. The doublet observables (S_x, S_y, S_z) and the inter-sheet phase $\psi_j = \angle(\text{HIGH}_j/\text{LOW}_j)$ are features absent from the DFT. For damage detection, one may test whether healthy structures produce stable doublet trajectories and whether damage perturbs (S_x, S_y, S_z); this remains an open validation problem against MAC criterion, SSI, PCA, and Mahalanobis indicators (see Table 1).

3. Classification support. Harmonic: $\Delta R^2 \approx 0.2$, $\Delta TV \approx 13$. φ -rational: $\Delta R^2 > 0.5$ for all classes tested (log-periodic, scale drift, Fibonacci drive). ΔTV is class-dependent: large for Fibonacci/braid-like signals (> 8), smaller for log-periodic (≈ 2). ΔR^2 is the more robust discriminant across all φ -structured types.

The FMT is not a DFT replacement

DFT is optimal for: harmonic signals, spectral compression, OFDM, standard frequency estimation.

FMT is the natural tool for: multiplicative lattices, log-periodic signals, discrete scale invariance, scale-drift tracking, quasi-periodic drives. Candidate domains (validation open): OMA/SHM log-scale coherence, fractal/self-affine textures.

Table 7: Scale-memory diagnostics (four signal types).

Signal	FMT TV	DFT TV	ΔTV	ΔR^2
Harmonic	0.035	13.17	13.14	0.199
Log-periodic	0.025	2.10	2.08	0.904
Scale drift	0.045	1.40	1.36	0.532
Fibonacci drive	0.020	8.97	8.95	0.929

References

- [1] P. Camelia, *Fibonacci-Autosimilar Compact Projection Maps: A Rank-Five SVD Characterization*, submitted to Linear Algebra Appl. (2026).
- [2] P.T. Dumitrescu et al., *Dynamical topological phase realized in a trapped-ion quantum simulator*, Nature **607**, 463–467 (2022).
- [3] S. Xu et al., *Non-Abelian braiding of Fibonacci anyons with a superconducting processor*, Nature Physics **20**, 1469–1475 (2024).
- [4] M.H. Freedman, A. Kitaev, Z. Wang, *Simulation of topological field theories by quantum computers*, Commun. Math. Phys. **227**, 587–603 (2002).
- [5] I. Daubechies, *Ten Lectures on Wavelets*, SIAM, Philadelphia (1992).
- [6] A. Córdoba, W. Gautschi, S. Ruscheweyh, *Vandermonde matrices on the unit circle*, Numer. Math. **57**, 577–591 (1990).
- [7] P. Camelia, *Amplitude-Free Scale-Ratio Disruption as a Gain-Invariant Validation Gate for Acoustic-Emission Monitoring of CFRP Failure*, applied companion paper (2026).
- [8] J. Bertrand, P. Bertrand, J.-P. Ovarlez, *The Mellin Transform*, in *The Transforms and Applications Handbook* (A.D. Poularikas, ed.), CRC Press (2000).
- [9] A. De Sena, D. Rocchesso, *A fast Mellin and scale transform*, EURASIP J. Adv. Signal Process. **2007**, 089170 (2007).
- [10] O. Christensen, *An Introduction to Frames and Riesz Bases*, 2nd ed., Birkhäuser (2016).

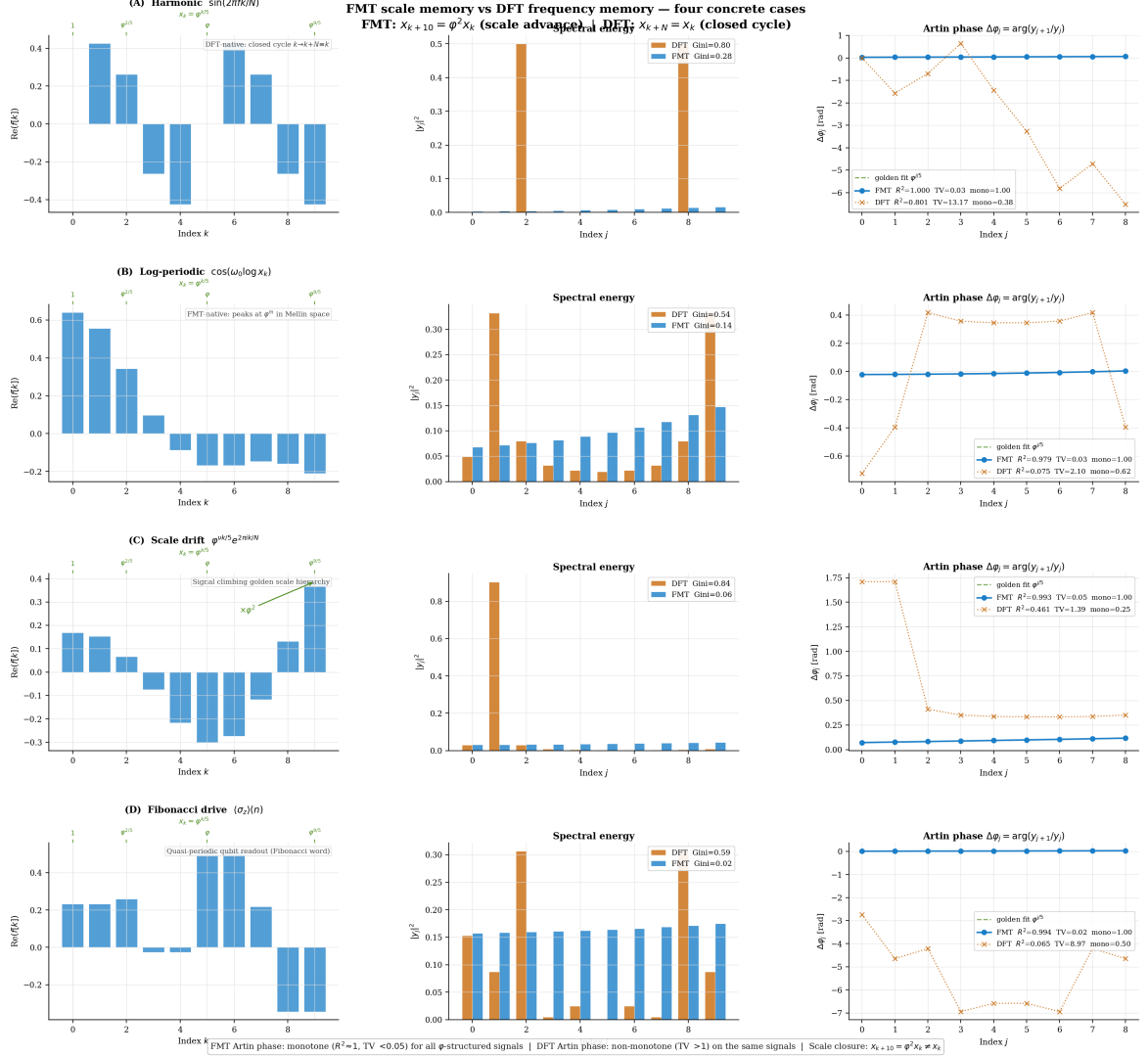


Figure 4: Scale memory demonstration (four signal types). Columns: signal on golden lattice $x_k = \varphi^{k/5}$; spectral energy FMT vs DFT (Gini); Artin phase $\Delta\phi_j$ vs golden target $\tau_j = \varphi^{j/5}$. Primary diagnostic: ΔTV and ΔR^2 (FMT–DFT contrast), not R^2_{FMT} alone.